# blacklistManagement

## Service Description

Service ID: *"blacklistManagement"*

**Abstract**
This document describes the `blacklistManagement` service, version 1.0.0, produced by the Blacklist support system of Arrowhead (documentation edition 5.3.0).

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# 1 Documentation edition and implementation scope

This is the 5.3.0 documentation edition, dated 2026-11-09. It carries forward the system and service contracts from the retained 5.2.2 documentation and identifies the current primary implementation sources. The former `documentation/v5.2.2` folder is retained unchanged.

Document version 5.3.0 is distinct from executable, dependency, registered system/service and wire-contract versions. Values still defined by the source (including 5.2.2 artefact versions, 1.0.0 service versions and `SysMonitor-AMQP-IDD-5.2.2` payload identifiers) are preserved. This documentation update does not change runtime behaviour or establish a new implementation release. Historical verification statements describe the tests recorded in the inherited baseline; this revision validates document compilation and references.

# 2 Overview

This document describes the `blacklistManagement` service, which provides operator management of the blacklist: adding systems with a reason and expiry, querying entries and inactivating them. It is produced by the Blacklist system [1] in service version 1.0.0.

The rest of this document is organized as follows. In Section 3, we describe the abstract message operations provided by the service. In Section 4, we end the document by presenting the data types used by the mentioned operations. The concrete interfaces are specified in the HTTP/TLS/JSON IDD [2] and the MQTT/TLS/JSON IDD [3].

## 2.1 Significant Prior Art

New in Arrowhead 5.

## 2.2 How This Service Is Meant to Be Used

Operators, the onboarding procedure (off-boarded systems are blacklisted) and security tooling add entries with `create`, inspect them with `query` and lift them with `remove`.

## 2.3 Important Delimitations and Dependencies

Entries are inactivated, not deleted, so the history is kept. Access follows the management policy of the Blacklist.

# 3 Service Interface

This section describes the interfaces to the `blacklistManagement` service. In particular, each subsection names an abstract operation, an input type and an output type, in that order. The input type is named inside parentheses, while the output type is preceded by a colon. Input and output types are only denoted when accepted or returned, respectively, by the interface in question.

All abstract data types named in this section are defined in Section 4.

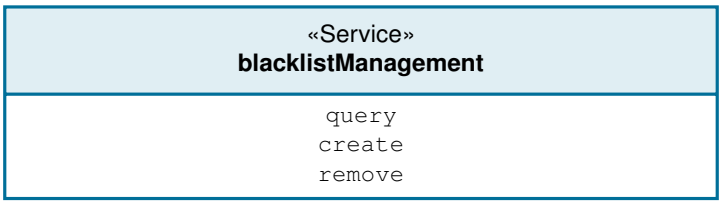



Figure 1: SysML-style block of the `blacklistManagement` service and its operations.

The following interface operations are available.

## 3.1 operation query (BlacklistQueryRequest) : BlacklistEntryListResponse

Returns a page of entries matching the filter.

## 3.2 operation create (BlacklistCreateListRequest) : BlacklistEntryListResponse

Adds blacklist entries with reason and optional expiry.

## 3.3 operation remove (List<Name>)

Inactivates the entries of the named systems.

# 4 Information Model

Here, all data objects that can be part of the `blacklistManagement` service are listed in alphabetic order. Note that each subsection, which describes one type of object, begins with the *struct* keyword, which is used to denote a collection of named fields, each with its own data type. As a complement to the explicitly defined types in this section, there is also a list of implicit primitive types in Section 4.7, which are used to represent things like hashes and identifiers.

## 4.1 struct BlacklistCreateListRequest

Entries to add to the blacklist.

| Field | Type | Description |
|---|---|---|
| entities | List<BlacklistCreateRequest> | Entries to create. |

## 4.2 struct BlacklistCreateRequest

One system to blacklist.

| Field | Type | Description |
|---|---|---|
| systemName | Name | System to blacklist. |
| expiresAt | DateTime | UTC expiry time; no value means no expiry. |
| reason | String | Reason of the blacklisting. |

## 4.3 struct BlacklistEntry

A blacklist entry.

| Field | Type | Description |
|---|---|---|
| systemName | Name | The blacklisted system. |
| createdBy | Name | System that created the entry. |
| revokedBy | String | System that inactivated the entry, if any. |
| createdAt | DateTime | UTC time of creation. |
| updatedAt | DateTime | UTC time of the last change. |
| reason | String | Reason of the blacklisting. |
| expiresAt | DateTime | Expiry of the entry; no value means permanent. |
| active | Boolean | Whether the entry is in force. |

## 4.4 struct BlacklistEntryListResponse

List of blacklist entries.

| Field | Type | Description |
|---|---|---|
| entries | List<BlacklistEntry> | Matching records of the current page. |
| count | Number | Total number of matching records (all pages). |

## 4.5 struct BlacklistQueryRequest

Paged filter for blacklist entries.

| Field | Type | Description |
|---|---|---|
| pagination | Page | Paging and sorting of the result. |
| systemNames | List<Name> | System names to match. |
| mode | String | Which entries to return: `ALL`, `ACTIVES` or `INACTIVES`. |
| issuers | List<String> | Systems that created the entries. |
| revokers | List<String> | Systems that inactivated the entries. |
| reason | String | Reason of the blacklisting. |
| alivesAt | DateTime | Only entries in force at this time. |

## 4.6 struct Page

Paging and sorting; all fields optional.

| Field | Type | Description |
|---|---|---|
| page | Number | Zero-based page number. |
| size | Number | Number of records per page (at most `max.page.size`, default 1000). |
| direction | String | Sort direction, `ASC` or `DESC`. |
| sortField | String | Field to sort by. |

## 4.7 Primitives

Types and structures mentioned throughout this document that are assumed to be available to implementations of this service. The concrete interpretations of each of these types and structures must be provided by any IDD document claiming to implement this service.

| Type | Description |
|---|---|
| Boolean | One out of `true` or `false`. |
| DateTime | Pinpoints a specific moment in time (UTC). |
| List | An *array* of a known number of items, each having type A. |
| Name | A string identifier that is intended to be both human and machine-readable. |
| Number | An integer number. |
| String | A chain of characters. |

# 5 References

[1] J. Delsing, “Blacklist – System Description,” Eclipse Arrowhead, SysD 5.3.0, 2026, ah5-Blacklist-java-spring/documentation/v5.3.0/Blacklist-v5.3.0-SysD.pdf.

[2] ——, “blacklistManagement HTTP/TLS/JSON – Interface Design Description (Blacklist),” Eclipse Arrowhead, IDD 5.3.0, 2026, blacklist-v5.3.0-blacklistManagement-HTTP-IDD.pdf.

[3] ——, “blacklistManagement MQTT/TLS/JSON – Interface Design Description (Blacklist),” Eclipse Arrowhead, IDD 5.3.0, 2026, blacklist-v5.3.0-blacklistManagement-MQTT-IDD.pdf.

# 6 Revision History

## 6.1 Amendments

| No. | Date | Version | Subject of Amendments | Author |
|---|---|---|---|---|
| 1 | 2026-10-22 | 5.2.2 | Document created from the Arrowhead LaTeX templates for the 5.2.2 Java Spring implementation, including the local extensions of this workspace. | Jerker Delsing |
| 2 | 2026-10-30 | 5.2.2 | Checked against the primary Java source tree (Eclipse-Arrowhead-java): services, operations, paths and data models unchanged. | Claude Opus 5.5 (draft) |
| 3 | 2026-11-09 | 5.3.0 | New documentation edition; primary-source references and Go/Rust implementation scope; 5.2.2 retained. | Documentation maintenance |

## 6.2 Quality Assurance

| No. | Date | Version | Approved by |
|---|---|---|---|
| 1 | | 5.3.0 | |